

SingleRAN

Intelligent Detection for Fronthaul Optical Paths Feature Parameter Description

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Contents

1 Change History.....

1.1 SRAN22.1 Draft A (2025-12-31).....

2 About This Document.....

2.1 General Statements.....

2.2 Applicable RAT.....

2.3 Differences.....

3 Overview.....

4 Intelligent Detection for Fronthaul Optical Paths.....

4.1 Principles.....

4.2 Network Analysis.....

4.2.1 Benefits.....

4.2.2 Impacts.....

4.3 Requirements.....

4.3.1 Licenses.....

4.3.2 Functions.....

4.3.2.1 Prerequisite Functions.....

4.3.2.2 Mutually Exclusive Functions.....

4.3.2.3 Function Impacts.....

4.3.3 Hardware.....

4.3.4 Networking.....

4.3.5 Cells.....

4.3.6 Others.....

4.4 Operation and Maintenance.....

4.4.1 Data Configuration.....

4.4.1.1 Data Preparation.....

4.4.1.2 Using MML Commands.....

4.4.1.3 Using the MAE-Deployment.....

4.4.2 Activation Verification.....

4.4.3 Network Monitoring.....

5 Parameters.....

6 Counters.....

1

1

3

3

3

4

5

7

7

10

10

10

10

11

11

11

11

11

13

14

16

16

16

16

16

17

19

19

24

25

27

7 Glossary.....

28

8 Reference Documents.....

29

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 04 (2025-08-30).

Technical Changes

Change Description	Parameter Change
Added link loss constraints for fronthaul networking using direct fiber connections. For details, see 4.3.4 Networking .	None
Added the description of triggering "EVT-26505 RF Unit CPRI Interface Switchover" in a ring or load sharing topology. For details, see 4.4.1.2 Using MML Commands .	None
Added impact relationships of optical path detection with RF module deep dormancy and RF module super dormancy in high-frequency NR TDD. For details, see 4.3.2.3 Function Impacts .	None

Editorial Changes

Added the description of fronthaul fault root cause diagnosis. For details, see [4.1 Principles](#).

Added the working principles of the OTDR based on Rayleigh backscattering and Fresnel reflection, and descriptions of the event dead zone and attenuation dead zone. For details, see [4.1 Principles](#).

Added the description of fronthaul fault root cause diagnosis. For details, see [4.1 Principles](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE and NR. There are no feature differences between them.

2.3 Differences

Differences Between NSA and SA

Function Name	Difference
Intelligent Detection for Fronthaul Optical Paths	None

Differences Between High Frequency Bands and Low Frequency Bands

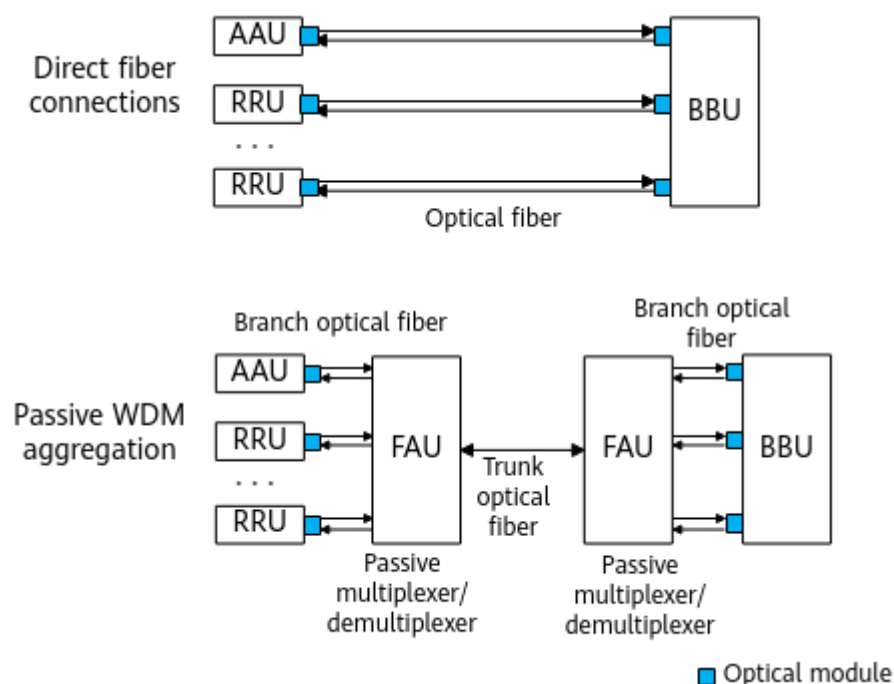
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference
Intelligent Detection for Fronthaul Optical Paths	None

3 Overview

In most cases, the fronthaul network for a base station is established using either direct fiber connections or passive WDM aggregation, as illustrated in [Figure 3-1](#). In fronthaul based on direct fiber connections, AAUs or RRUs are directly connected to a BBU, with each fronthaul link operating independently. In fronthaul based on passive WDM aggregation, data from multiple branch optical fibers is aggregated onto one trunk optical fiber. This fronthaul solution requires passive multiplexer/demultiplexer modules, such as Huawei fronthaul aggregation units (FAUs).

Figure 3-1 Direct fiber connections and passive WDM aggregation

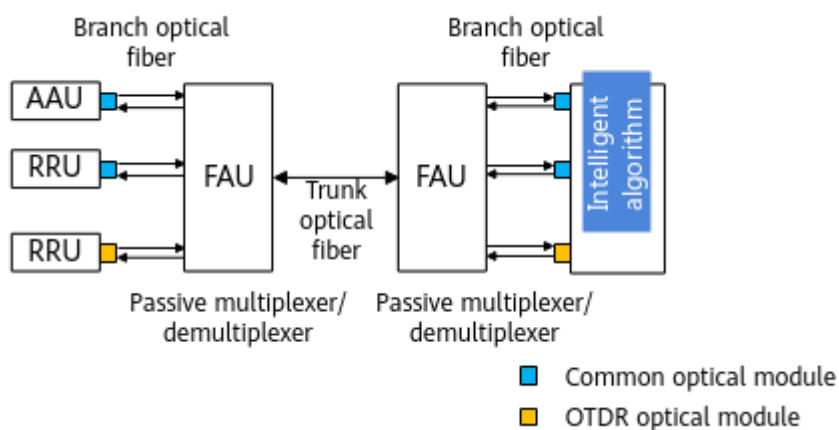


In traditional detection of optical path faults, maintenance personnel visit a site to locate faults based on alarms reported from a base station. They need remote assistance from engineers who are not onsite, and specific tools, such as optical power meters and pen-style visual fault locators (VFLs). This detection method demands highly experienced maintenance personnel and is both time- and labor-

intensive. It often requires multiple site visits, leading to low efficiency. Another solution is to use the optical time domain reflectometer (OTDR) to accurately locate optical path faults. However, OTDRs are typically expensive and require highly skilled personnel to operate.

To address this issue, Huawei introduces Intelligent Detection for Fronthaul Optical Paths. **Figure 3-2** shows how this function works. Using OTDR-capable optical modules (or OTDR optical modules) and intelligent algorithms, this function remotely detects faults on fronthaul optical paths, accurately locates them, and identifies their causes. This significantly improves fault detection efficiency. This function applies to macro and LampSite base stations. In LampSite base stations, this function works only for the downlink optical path from the BBU to the RHUB using direct fiber connections.

Figure 3-2 Intelligent Detection for Fronthaul Optical Paths (using macro base stations with passive WDM aggregation as an example)



4 Intelligent Detection for Fronthaul Optical Paths

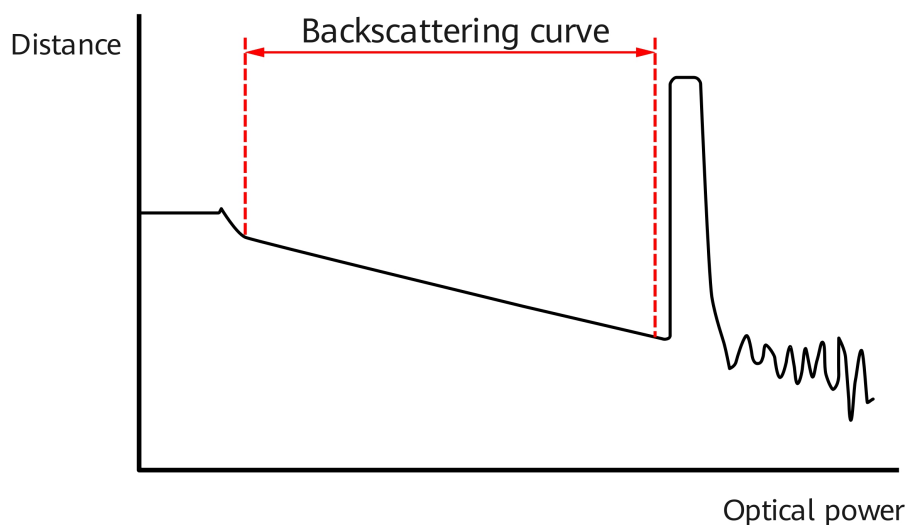
4.1 Principles

With intelligent detection for fronthaul optical paths, OTDR optical modules transmit specific pulse signals to optical fibers and analyze Rayleigh backscattering and Fresnel reflection results to determine the physical characteristics (such as length and signal loss) of given optical paths. An intelligent algorithm then processes the data to accurately locate optical path faults and identify their causes. Optical path detection will interrupt ongoing fronthaul services because the specific pulse signals from OTDR optical modules are incompatible with service signals on fronthaul optical paths.

The OTDR works based on Rayleigh backscattering and Fresnel reflection.

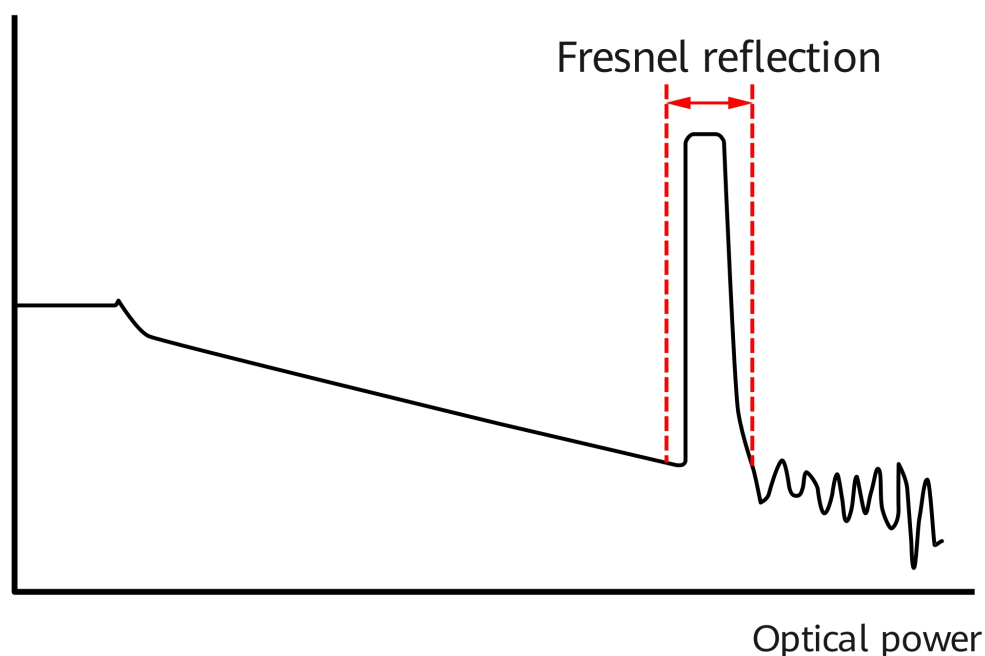
- Rayleigh backscattering: When an OTDR signal encounters an irregularity in the fiber, part of the optical power is scattered in multiple directions. This phenomenon is known as Rayleigh scattering. When a portion of the scattered optical power returns toward the light source, it is referred to as Rayleigh backscattering, and the returned optical power is called backward-scattered light. Due to scattering loss, the intensity of backward-scattered light pulses decreases significantly. [Figure 4-1](#) shows the backscattering curve.

Figure 4-1 Backscattering curve



- **Fresnel reflection:** Fresnel reflection occurs at the boundary between two transmission media with different refractive indices. When light encounters such a boundary, a portion of the signal is reflected back toward the OTDR module. The module then calculates the distance to the boundary point based on the echo time, which forms the fundamental ranging principle of the OTDR. On the OTDR curve, these reflections appear as sharp "spike" peaks, known as reflective events. [Figure 4-2](#) shows the Fresnel reflection curve.

Figure 4-2 Fresnel reflection curve

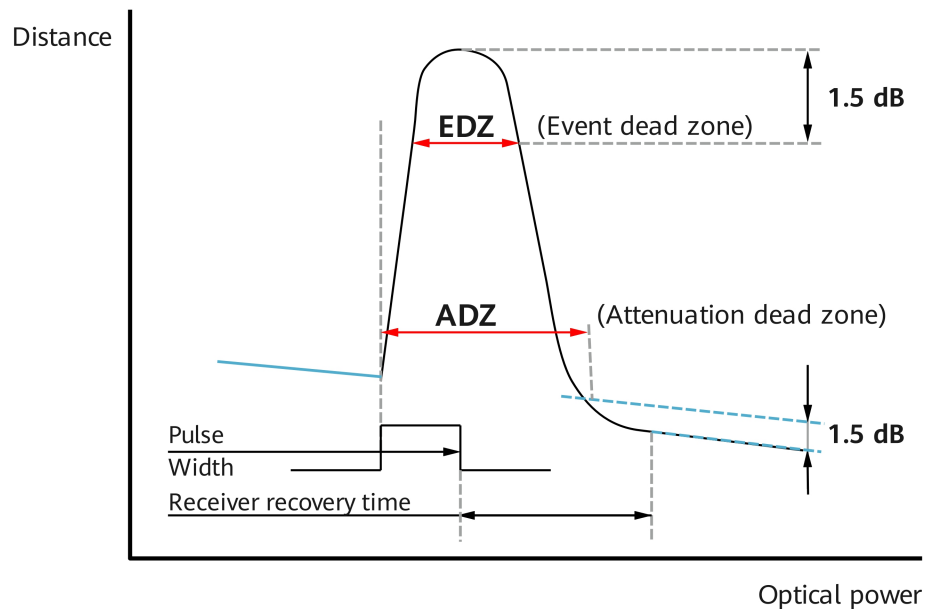


On the optical path near the optical module, any event dead zone or attenuation dead zone occurs because the egress of the optical module is not seamlessly aligned with the end face of the optical fiber. As a result, the detection precision

within the dead zone decreases. **Figure 4-3** shows the relationship between the optical power and the distance between reflective events.

- Event dead zone: the distance between two cursors set 1.5 dB below a reflective peak. It can also be described as the minimum distance between two reflective events at which an accurate length measurement can be made. Within the dead zone, the distance to each event can be measured, but the loss of each event cannot. Detection accuracy returns to normal once the distance is beyond the dead zone.
- Attenuation dead zone: the distance between the rising edge of a pulse and the point where the backscatter level deviates by 0.5 dB from a straight-line fit. It can also be described as the minimum distance between two reflective events at which an accurate loss measurement can be made. Detection accuracy returns to normal once the distance is beyond the dead zone.

Figure 4-3 Relationship between the optical power and the distance between reflective events



With OTDR-based fronthaul fault root cause diagnosis in effect, an OTDR detection is performed and a fronthaul fault root cause alarm is reported based on the detected event location when the CPRI port of the OTDR optical module is faulty and services are unavailable. For details, see section "OTDR-based Fronthaul Fault Root Cause Diagnosis" in *Fault Management*.

OTDR optical modules can transmit service signals and detect optical paths. You can run the **STR OPTPATHQUALITYDET** command to start the optical path detection function. Before optical path detection starts, OTDR optical modules transmit service signals. After optical path detection starts, OTDR optical modules transmit specific pulse signals for optical path detection, and then receive and process the returned signals. After this, an intelligent algorithm analyzes both uplink and downlink optical paths, including the status of the optical fibers, multiplexer/demultiplexer modules, and connectors, to determine whether a fault occurs. After optical path detection is complete, the base station reports the detection result. **DSP OPTPATHQUALITYDET** can be executed to query the

detection result and **STP OPTPATHQUALITYDET** to stop a detection task. A maximum of five optical path detection tasks can be performed on a single NE at the same time.

With OTDR-based fronthaul fault root cause diagnosis in effect, an OTDR detection is performed and a fronthaul fault root cause alarm is reported based on the detected event location when the CPRI port of the OTDR optical module is faulty and services are unavailable. For details, see section "OTDR-based Fronthaul Fault Root Cause Diagnosis" in *Fault Management*.

4.2 Network Analysis

4.2.1 Benefits

The function is primarily used for remote detection for fronthaul optical paths during site deployment and maintenance, enabling efficient identification of optical path faults.

4.2.2 Impacts

- The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).
- This function interrupts all services carried on the fronthaul links of the base station. Therefore, this function is limited to only the following scenarios:
 - Site deployment: This function can be used to locate optical path faults in the engineering phase.
 - Maintenance: This function can be used to locate optical path faults that cause cell unavailability or fronthaul link interruption.

NOTICE

This function interrupts the fronthaul services of the base station and cannot be used when cells are working properly.

- Starting optical path detection may cause the following alarms:
 - ALM-26234 BBU CPRI Interface Error
 - ALM-26324 BBU IR Interface Error
 - ALM-26235 RF Unit Maintenance Link Failure
 - Cell Unavailable
 - Cell TRP Capability Degraded
- This function increases CPU usage of baseband processing units and RF modules. Excessive CPU usage may affect user access to services on ports other than the optical path detection port of the baseband processing units.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

- During optical path detection, the following RF module logs cannot be exported: **RRU logs**, **RF alarm logs**, **RF temperature logs**, and **remote simplified baseband logs**. Optical path detection is not allowed when RF module logs are being exported. You are advised to start optical path detection after RF module logs are exported.
- Execution of a certain MML command times out or fails if (1) the execution occurs either before optical path detection starts or during such detection and (2) the command (such as **STR HWOLTST** or **DSP ELABEL**) involves interaction with an RF module.
- Optical path detection affects certain energy saving functions, as listed in the following table.

NOTE

- If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state.
- If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies:
 - In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process starts. Once 30 minutes elapse, they will evaluate whether to re-enter the dormancy state.
 - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.

RAT	Function Name	Function Switch	Reference
LTE FDD LTE TDD NR FDD NR TDD (low frequency) NR TDD (high frequency)	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>
LTE FDD LTE TDD NR FDD NR TDD (low frequency) NR TDD (high frequency)	RF module super dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>
LTE FDD LTE TDD NR	pRRU Deep Dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON	<i>Multi-RAT Coordinated Energy Saving</i>

RAT	Function Name	Function Switch	Reference
LTE FDD LTE TDD NR FDD NR TDD (low frequency)	Adaptive BBU deep dormancy	- LTE TDD: TDD_BBP_DEEP_DORMANCY_SW option of the ENodebPwrSavingAlgo.DormancySwitch parameter - LTE FDD: FDD_BBP_DEEP_DORMANCY_SW option of the ENodebPwrSavingAlgo.DormancySwitch parameter - NR: BBP_DEEP_DORMANCY_SW option of the gNB PwrSavingAlgo.DormancySwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

DBS3900 LampSite and DBS5900 LampSite

BBUs

BBU3900, BBU3910, BBU5900, BBU5910

Boards

- Main control boards: UMPTb, UMPTe, UMPTg, UMPTga, and UMPTHa series boards
- Baseband processing units: UBBPd, UBBPe (except UBBPeb), UBBPg, and UBBPh series boards that support a CPRI port rate of 10 Gbit/s, 25 Gbit/s, or 50 Gbit/s

RF Modules

AAUs and RRUs must support a CPRI port rate of 25 Gbit/s or 50 Gbit/s and the OTDR function. To check whether the OTDR function is supported, run the **DSP RRU** or **DSP CXU** command and check the **OPTICAL_PATH_QUALITY_DETECTION** setting of the **Function Support** field in the command output.

Optical Modules

- Optical modules that transmit optical path detection signals must support the OTDR function. (To check whether the OTDR function is supported, run the **DSP SFP** command and check the **OPTICAL_PATH_QUALITY_DETECTION** setting of the **Function Support** field in the command output.)

NOTE

Unlike common optical modules, OTDR optical modules have the "OTDR" marking on both paper labels and electronic labels (accessible using **DSP ELABEL**).

- The optical module type is SFP.
- The operating wavelength of OTDR optical modules is 1331 nm or 1271 nm.
- If fronthaul is established using direct fiber connections or passive WDM aggregation, the optical module configuration must meet any of the following requirements:
 - The BBU uses an OTDR optical module operating at a wavelength of 1331 nm, and the RHUB uses a common optical module (its operating wavelength depends on networking requirements).
 - The BBU uses an OTDR optical module operating at a wavelength of 1331 nm, and the AAU uses a common optical module (its operating wavelength depends on networking requirements).
 - The AAU uses an OTDR optical module operating at a wavelength of 1271 nm, and the BBU uses a common optical module (its operating wavelength depends on networking requirements).
 - The BBU uses an OTDR optical module operating at a wavelength of 1331 nm, and the AAU uses an OTDR optical module operating at a wavelength of 1271 nm.

NOTE

If an OTDR optical module is used only at one end of the optical path and a common optical module is used at the other end, only the unidirectional optical path detection from the OTDR optical module to the common optical module is supported.

- An OTDR optical module supports a rate of 25 Gbit/s or 50 Gbit/s, and a common optical module supports a rate of 10 Gbit/s, 25 Gbit/s, or 50 Gbit/s (10 Gbit/s common optical modules are limited to fronthaul networking using passive WDM aggregation).

4.3.4 Networking

- This function applies to fronthaul networking using direct fiber connections or passive WDM aggregation. (Optical path detection between a BBU and an RHUB is supported only when direct fiber connections are used.)
 - For direct fiber connections, only two-fiber bidirectional networking is supported; single-fiber bidirectional (BIDI) networking is not supported.
 - For direct fiber connections, if a 25 Gbit/s OTDR optical module is used at one end of the link and a 25 Gbit/s common optical module is used at the other end, the link loss must be greater than 4 dB; If a 50 Gbit/s OTDR optical module is used at one end of the link and a 50 Gbit/s common optical module is used at the other end, the link loss must be greater than 2.5 dB. The link loss can be queried by running the **DSP SFP**

command (with the **SFP.QRYINFOTYPE (LTE eNodeB, 5G gNodeB)** parameter set to **LINK**).

- For passive WDM aggregation, the CPRI ports on the optical modules of the baseband processing unit must be next to each other, and the optical modules must connect directly to the multiplexers/demultiplexers.
- For passive WDM aggregation, this function is limited to the following scenarios.

Transmission Wavelength on the BBU (nm)	Transmission Wavelength on the AAU/RRU (nm)
1331	1271
1351	1291
1371	1311

- For direct fiber connections, if the BBU uses an OTDR optical module operating at a wavelength of 1331 nm and the AAU uses a common optical module, or if the AAU uses an OTDR optical module operating at a wavelength of 1271 nm and the BBU uses a common optical module (the typical wavelength is 1310 nm), the following link loss constraints apply:
 - Avoid connecting a 10 Gbit/s common optical module with an OTDR optical module in this networking scenario. You are advised to deploy the OTDR optical module operating at a wavelength of 1331 nm on the BBU side and the OTDR optical module operating at a wavelength of 1271 nm on the AAU side. Ensure that the link loss is greater than 3 dB. The link loss can be queried by running the **DSP SFP** command (with the **SFP.QRYINFOTYPE (LTE eNodeB, 5G gNodeB)** parameter set to **LINK**).
 - If a 25 Gbit/s common optical module is used, ensure that the link loss is greater than 4 dB. The link loss can be queried by running the **DSP SFP** command (with the **SFP.QRYINFOTYPE (LTE eNodeB, 5G gNodeB)** parameter set to **LINK**).
 - If a 50 Gbit/s common optical module is used, ensure that the link loss is greater than 3.5 dB. The link loss can be queried by running the **DSP SFP** command (with the **SFP.QRYINFOTYPE (LTE eNodeB, 5G gNodeB)** parameter set to **LINK**).
- The fronthaul transmission distance must not exceed 10 km (as the maximum optical path detection range is limited to 10 km). A link should have no more than four connectors (one fiber distribution terminal, or FDT, counted as a single connector), excluding the multiplexer/demultiplexer.
- The star, chain, ring, load sharing, trunk chain, and dual-star topologies are supported.
 - In chain and ring topologies, link detection is supported only between the BBUs and RF modules, but not between RF modules.
 - In ring, trunk chain, and load sharing topologies, uplink optical path detection cannot be started simultaneously for two ports on an RF module.
- If the CPRI port connection information between the baseband processing unit and an RF module cannot be queried (in the **DSP RRCHAINPHYTOPO**

command output), for example, due to the RF module being disconnected, uplink optical path detection cannot be performed for that RF module.

4.3.5 Cells

None

4.3.6 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

- [Table 4-1](#) lists the parameters that need to be set in the **STR OPTPATHQUALITYDET** command for starting optical path quality detection.

Table 4-1 Parameters for starting optical path quality detection

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CPRICN	Set this parameter based on the network plan.
Subrack No.	CPRISRN	Set this parameter based on the network plan.
Slot No.	CPRISN	Set this parameter based on the network plan.
Port No.	PN	Set this parameter based on the network plan.
Detect Direction	DETECTDIRECTION	Set this parameter based on the network plan.
Force Execute	FORCEEXECUTE	You are advised to set this parameter to NO to prevent the mistaken start of optical path quality detection, as such unintended detection affects services.

 NOTE

- The **Cabinet No.**, **Subrack No.**, **Slot No.**, and **Port No.** parameters in the **STR OPTPATHQUALITYDET** command must be set based on the cabinet number, subrack number, slot number, and port number of the OTDR optical module on the BBU, respectively. **Detect Direction** uses the BBU as the start node for a detection process. The value **DL** indicates detection from the BBU to the RF module, and **UL** indicates detection from the RF module to the BBU.
- Starting optical path quality detection will interrupt all services carried on the fronthaul link of the base station and cause link- and cell-related alarms. Therefore, you are advised to start the detection only when a link or cell fault occurs. If certain issues (such as bit errors and cell KPI deterioration) persist despite a normal CPRI link, you need to start optical path quality detection (by setting **FORCEEXECUTE** to **YES**) to locate optical path faults.
- [Table 4-2](#) lists the parameters that need to be set in the **STP OPTPATHQUALITYDET** command for stopping optical path quality detection.

Table 4-2 Parameters for stopping optical path quality detection

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CPRICN	Set this parameter based on the network plan.
Subrack No.	CPRISRN	Set this parameter based on the network plan.
Slot No.	CPRISN	Set this parameter based on the network plan.
Port No.	PN	Set this parameter based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- Starting optical path quality detection will interrupt the fronthaul services of the base station. Therefore, start the detection when the fronthaul link is faulty and the cell is unavailable or when the fronthaul link is disconnected. Do not start the detection when the cell is working properly.
- When optical path quality detection is successfully started, "EVT-26407 Fronthaul optical path quality detection start" is reported, and then link- and cell-related alarms are generated. After optical path quality detection is completed or stopped, "EVT-26408 Fronthaul optical path quality detection stop" is reported. Since the effects of the detection task do not immediately dissipate, link- and cell-related alarms may continue to be generated and cleared within several minutes after the detection task ends.
- During optical path quality detection, the cell services carried on the detection link are unavailable. If you run the **DSP CELL**, **DSP NRDUCELL**, or **DSP NRDUCELLTRP** command to query the status of the corresponding cell, the value of **Cell Latest State Change Reason**, **NR DU Cell Latest State Change Reason**, or **NR DU Cell TRP Unavailable Reason** is **Fronthaul optical path quality detection is in progress** in the command output.
- In CPRI MUX scenarios, only the link between the converging board and the RF module can be detected. The corresponding start and stop events for the detection process are reported exclusively on the converging party. Additionally, only the converging party will display the cell status change cause/cell unavailability cause **Fronthaul optical path quality detection is in progress**. For the meanings of the converging board and converging party, see *CPRI MUX*.
- If an optical path quality detection task causes connection failures for the boards on the involved RRU chain/ring or prevents established links from maintaining the maximum CPRI line rate, you need to manually initiate a negotiation by running the **STR CPRI LBRNEG** command.
- A maximum of five optical path quality detection tasks can be performed on a single NE at the same time.
- A single optical path quality detection task can be completed within 11 minutes. This means that fronthaul is interrupted for less than 11 minutes.
- EVT-26326 Optical Module Rate Mismatch will not be reported when 25 Gbit/s OTDR optical modules are used on BBUs and 10 Gbit/s common optical modules are used on the peer side.
- In LampSite scenarios, only the downlink optical path from the BBU to the RHUB can be detected (the BBU uses an OTDR optical module and the RHUB uses a common optical module).
- In a ring or load sharing topology, enabling OTDR detection on the uplink CPRI link will cause link disconnection. The RF unit will then automatically switch to the downlink CPRI link, triggering the "EVT-26505 RF Unit CPRI Interface Switchover" event.

Activation Command Examples

1. Check that optical modules support the function. (Check the **OPTICAL_PATH_QUALITY_DETECTION** setting of the **Function Support** field in the command output.)

```
//Querying the dynamic information about an optical or electrical module (Cabinet No.: 0; Subrack  
No.: 0; Slot No.: 1; Port Type: CPRI; Port No.: 0):  
DSP SFP: QRYINFOTYPE=MODULE, CN=0, SRN=0, SN=1, PT=CPRI, MODULEID=0;
```

2. Start optical path quality detection.

```
//Starting optical path quality detection (Cabinet No.: 0; Subrack No.: 0; Slot No.: 1; Port No.: 0):  
STR OPTPATHQUALITYDET: CPRICN=0, CPRISRN=0, CPRISN=1, PN=0;
```

Deactivation Command Examples

Stop optical path quality detection.

```
//Stopping optical path quality detection (Cabinet No.: 0; Subrack No.: 0; Slot No.: 1; Port No.: 0):  
STP OPTPATHQUALITYDET: CPRICN=0, CPRISRN=0, CPRISN=1, PN=0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- If the **STR OPTPATHQUALITYDET** command is executed successfully, the detection progress is automatically reported. After optical path quality detection is complete, **Optical Path Quality Detection Result** and **Optical Path Quality Detection Details** are displayed. You can also run the **DSP OPTPATHQUALITYDET** command to manually query the information. The optical path quality detection results are as follows:
 - a. If the detection is not started, **Up Link Detect Result** or **Down Link Detect Result** in the **Optical Path Quality Detection Result** area is **USER_NOT_START**.
 - b. If the detection is successful, **Up Link Detect Result** or **Down Link Detect Result** in the **Optical Path Quality Detection Result** area is **SUCCESS**, and **Optical Path Quality Detection Details** is displayed. The following table describes the fields in the **Optical Path Quality Detection Details** area.

Table 4-3 Fields in the Optical Path Quality Detection Details area

Field	Description
Optical Path Node Type	Indicates the possible type of a node over an optical path. The value can be TRANSMIT_END, CONNECTOR, MUX, CONNECTOR_OR_MUX, FIBER, or FINAL_DETECT_POINT. • The value TRANSMIT_END indicates the port over which the beam used in optical path quality detection is transmitted. • The value CONNECTOR_OR_MUX indicates that the type of the node may be a multiplexer/demultiplexer or connector. • The value FINAL_DETECT_POINT indicates the last node discovered in the optical path quality detection.
Fault Type	Indicates the fault type corresponding to a node over an optical path. The value can be DIRTY_CONNECTOR, BENT_FIBER, OPTICAL_PATH_INTERRUPTED, OPTICAL_PATH_END, or NORMAL. If the value of this parameter is OPTICAL_PATH_INTERRUPTED, a fault occurs on the optical path node, causing optical signals to be interrupted at this node. If the value of this parameter is OPTICAL_PATH_END, optical signals are interrupted at the optical path node; in this case, if other optical path nodes are normal while the CPRI link is still abnormal, you are advised to check the optical path node.
Distance (m)	Indicates the distance between a node over an optical path and the optical module. The results in Downlink Optical Path Quality Detection Details indicate the distances between nodes over the optical path and the optical module on the baseband processing unit. The results in Uplink Optical Path Quality Detection Details indicate the distances between nodes over the optical path and the optical module on the RF module. The difference between the value of this parameter and the actual distance is less than 10 m.
Reflectivity (0.1dB)	Indicates the reflectivity of a node over an optical path. If the reflectivity is 0, the node does not generate a reflection signal.
Loss (0.1dB)	Indicates the loss of a node over an optical path. When the value of the Optical Path Node Type parameter is FINAL_DETECT_POINT for a node, the value of the Loss (0.1dB) parameter corresponding to this node is invalid and displayed as NULL.

 NOTE

- Physical differences, such as those in types, contamination levels, and end face flatness, may exist between connectors of uplink and downlink nodes. These differences can affect the reflectivity of an optical signal, leading to varying detection results for the uplink and downlink.
 - If two or more connectors on a fronthaul optical path are severely contaminated (for example, with reflectivity exceeding -15 dB), the fault location may not be accurately identified using topology-based analysis.
 - If a node over an optical path is saturated (meaning that the light reflected from the node is beyond the receiving capability of the optical module), other nodes that are beyond a certain distance from this saturated node may fail to be identified during optical path detection. If the saturated node is **FINAL_DETECT_POINT**, the detected distance between the node and the optical module may exceed the actual distance.
 - According to IEEE standards and construction specifications, the typical reflectivity for a standard connector should be less than or equal to -35 dB, with a typical loss of less than or equal to 0.5 dB. For a multiplexer/demultiplexer, the typical reflectivity should also be less than or equal to -35 dB, with a typical loss of less than or equal to 2.3 dB. You can preliminarily determine whether the connector and multiplexer/demultiplexer nodes are dirty based on these typical values. If the type of a node over an optical path is **CONNECTOR_OR_MUX**, you are advised to determine whether the node is dirty based on the typical reflectivity and loss of a multiplexer/demultiplexer.
 - If the actual distance between two nodes over an optical path is less than 20 m, optical path quality detection may mistakenly identify the two nodes as a single node.
 - Optical path quality detection cannot accurately detect exceptions in the area about 30 m (called the transmit-end dead zone) beyond the detection start point.
 - **FINAL_DETECT_POINT** indicates the final position that is actually detected, which may be the peer optical module or a position on the optical path (optical path quality detection is terminated in advance due to factors such as optical path interruption and excessive optical path attenuation). If the reported reflectivity of **FINAL_DETECT_POINT** is 0, the final detection point is on the optical path. If bidirectional optical path quality detection is enabled, the final detection point can be determined based on the detection results of the uplink and downlink optical paths.
 - If the reported optical path fault has been resolved and all optical path nodes are back to normal status, but link alarms and exceptions persist, you are advised to check for potential issues such as: transmit-end dead zone exceptions, final detection point exceptions, optical module exceptions, BBU/RF module exceptions, and exceptions in optical paths other than the currently detected optical path.
 - **Distances for the Latest Optical Path Quality Detection** indicates the optical fiber length detected during the latest optical path quality detection. This parameter serves as a reference for interpreting the detection results. If the current optical path quality detection result shows that all optical paths are normal but the CPRI link remains abnormal, and the final detection point in the current result is shorter than the distance recorded in the latest detection, you are advised to check the final detection point.
- c. If the detection fails, **Up Link Detect Result** or **Down Link Detect Result** in the **Optical Path Quality Detection Result** area is **FAILURE**. Rectify the fault based on **Up Link Detect Fail Reason** or **Down Link Detect Fail Reason**. The following table provides the details.

Result	Description
STARTUP_FAIL	When the detection starts, the baseband processing unit or RF module does not respond or fails to respond.
DETECT_FAIL	During the detection, several issues may cause the task to receive no response or fail to get a response. These issues include: The optical module is removed and inserted; the baseband processing unit or RRU does not respond or fails to respond; and the detection data is abnormal.
OPTICAL_MODULE_AB NORMAL	The optical module is abnormal. Check whether an alarm related to the optical module is generated.
RF_MODULE_NOT_SU PPORT	The RF module does not support this function.
RF_MODULE_NOT_CO NFIGURED	The RF module has not been configured.
RF_LINK_UNAVAILAB LE	The RF maintenance link is not set up.
OPTICAL_MODULE_N OT_SUPPORT	The optical module does not support this function.
BOARD_UNAVAILABLE	Availability Status of the RF module is not Normal .
PEER_OPT_MODULE_S END_SIG	The optical module on the RF module side is transmitting optical signals, which affects optical path quality detection. The possible cause is an incorrect connection to the TX and RX ports on the optical module.
UP_LINK_RX_NOSIGNA L	The optical module at the uplink receive end does not receive signals, indicating that a fault occurs on the uplink. In this case, you are advised to check the uplink first.

- If the **STR OPTPATHQUALITYDET** command fails to be executed, rectify the fault based on the displayed error message. The following table lists the error messages.

Scen ario	Error Message	Description
1	The current main control board does not support this function.	The main control board in use does not support this function.

Scenario	Error Message	Description
2	This function is not supported in the current networking scenario.	Optical path detection is not supported in certain networking scenarios, including connections between RHUBs and pRRUs, direct connections between hRRUs/mRRUs and the DCU, direct connections between hRRUs/mRRUs and the BBU, and connections between RRUs.
3	The DSP BRDDORMANCY command can be executed to query whether a board is in the dormancy state. If a board is in the timing dormancy state or is exiting any other dormancy state, perform this operation after it exits the state.	The baseband processing unit or RF module is in the dormancy state (as indicated in the DSP BRDDORMANCY command output). You must wait until it exits the dormancy state before running the STR OPTPATHQUALITYDET command.
4	The board is unavailable.	The baseband processing unit is not configured or in position, or is not working.
5	The board has not been configured.	The baseband processing unit has not been configured.
6	The board does not support this function.	The baseband processing unit does not support this function.
7	Invalid port number.	The port number is out of the range supported by the baseband processing unit.
8	The optical module of the specified port is not in position.	No optical module is available in the specified port.
9	The number of test instances reaches the upper limit.	The number of instances for which optical path quality detection has been started reaches the maximum.
10	The test is already started.	Optical path quality detection has been started on the port. Do not start it again.
11	Optical path quality detection can be enabled for only one port on a board at a time.	Optical path quality detection has been started for a port on a baseband processing unit. Detection for other ports can be started only after the ongoing detection process is complete.

Scen ario	Error Message	Description
12	The laser switch of the optical module is off.	The laser switch of the port is turned off and optical path quality detection cannot be started.

4.4.3 Network Monitoring

To monitor and analyze the latest optical path status, export the optical path quality detection logs (recording optical path detection data and detection results). To export such logs, run the **ULD FILE** command with **Source File Type** set to **OTDRLOG**.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *Multi-RAT Coordinated Energy Saving*
- *CPRI MUX*
- *Fault Management*